Data Mining: Assignment Week 0

1. A software consists of a collection of -

A. programs

B. circuits

C. humans

B. network

2. A high level program is converted to machine language code using -

A. compiler

B. hardware

C. network

D. hard-disk

3. Which of the following is NOT a data structure?

A. linear search

B. binary tree

C. linked list

D. stack

4. Access to which of the following memory is slowest?

A. CD-ROM

B. Hard Disk

C. RAM

D. Magnetic Tape

5. The worst-case complexity of sorting an integer array having *n* elements using the bubble sort algorithm is:

A. n

B. n^2

C. n^3

D. n^4

6. Which of the following database models primarily uses tables to represent data?

A. relational

B. semi-structured

C. file based

D. unstructured

7. The maximum value of the function $f(x) = -x^2 + 4x + 1$ is?

A. 2

B. 3

C. 4

D. 5

8. First derivative $f'(x)$ with respect to x of the function $f(x) = 3x^4 - 5x^3 + 2x^2 - 7x + 10$ is?

A. $3x^3 - 5x^2 + 2x - 7$

B. $5x^3 + 2x^2 - 7x$

C. $3x^2 - 5x + 10$

D. $3x^4 - 5x^3 + 2x^2 + 10$

9. Which of the following in a two-dimensional plane is represented by the equation $b_1x_1 + b_2x_2 + a_0 = 0$

A. a straight line

B. a parabola

C. a hyperbola

D. a circle

10. Median of the set of numbers {2, 4, 6, 8, 10} is?

A. 6

B. 2

C. 8

D. 4

11. Mean of the set of numbers {2, 4, 6, 8, 10} is?

A. 2

B. 4

C. 6

D. 8

Data Mining: Assignment Week 1

1. Data mining is the process of finding valid, novel, useful, and actionable, _____ in large volume of data. Which of the following terms best fills the gap above?

A. volume

B. errors

C. patterns

D. noise

2. Which of the following is usually the first step in the data mining process?

A. Visualization

B. Preprocessing

C. Modeling

D. Deployment

3. "Country name", can be considered as an attribute of type?

A. Nominal

B. Ordinal

C. Interval

D. Ratio

4. Average per capita income of a citizen of country, can be considered as an attribute of type?

A. Nominal

B. Ordinal

C. Interval

D. Ratio

5. Which of the following operations is allowed to be performed on nominal attributes?

A. Ranking

B. Addition

C. Multiplication

D. Distinctness

6. Which of the following operations *cannot* be performed on ordinal attributes?

A. Distinctness

B. Multiplication

C. Ordering

D. Median finding

7. List of grades obtained by a student and stored in university database, is an example of -

A. Record data

B. Tree data

C. Graph data

D. Stacked data

8. Attributes of data instances in a data table is represented as?

A. Blank spaces

B. Columns

C. Rows

D. Delimiters

9. Which of the following is an example of text data?

A. Weight of a person

B. Height of a person

C. Gender of a person

D. Biodata description of a person

10. A graph data is represented as a set of:

A. Row vectors

B. Nodes and edges

C. Column vectors

D. Text strings

11. Which of the following action is performed to reduce the number of rows in a data table:

A. Data sampling

B. Dimensionality reduction

C. Noise cleaning

D. Discretization

Data Mining: Assignment Week 2

1. NPTEL offers a total of N number of courses. The number of possible combinations of any number (excluding zero) of courses that a student can take is:

A. N^2-1

B. 2^N-1

C. N/2

D. 2N

2. Say, NPTEL offers total 7 courses. A student wants to take a set of 3 courses. The possible number of choices are:

A. 75

B. 55

C. 35

D. 15

3. We have an association rule of the form $X \rightarrow Y$, where X and Y are itemsets in a supermarket. The fraction of transactions which contain both X and Y is known as:

A. Support

B. Confidence

C. Accuracy

D. Perspicuity

4. A frequent itemset is one which satisfies the -

A. Support criteria

B. Confidence criteria

C. Accuracy criteria

D. Perspicuity criteria

5. Milk and Honey are two items that are sold in a supermarket. Consider the itemsets A = {Milk} and the

itemset B = {Milk, Honey}. Which of the following statement is *always* true?

A. support(A) ≤ support(B)

B. **support(A) ≥ support(B)**

C. support(A) = support(B)

D. support(A) ≠ support(B)

6. Consider three itemsets X = {milk, honey, bread}, Y = {milk, honey}, Z = {milk}. Which of the following statements are correct?

A. support(X) > support(Y)

B. support(Z) > support(Y)

C. support(X) > support(Y)

D. support(Y) > support(Z)

7. In the following data table, what is the support of the itemset {b, c, d}?

Transaction ID	Itemsets
1	{a, b, d, e}
2	{b, c, d}
3	{a, b, d, e}
4	{a, c, d, e}
5	{b, c, d, e}
6	{b, d, c}
7	{c, d}
8	{a, b, c}
9	{a, d, e}
10	{b, c}

A. 0.1

B. 0.3

C. 0.5

D. 0.7

8. In the following data table, what is the confidence of the rule a → b?

Transaction ID	Itemsets
1	{a, b, d, e}

2	{b, c, d}
3	{a, b, d, e}
4	{a, c, d, e}
5	{b, c, d, e}
6	{b, d, c}
7	{c, d}
8	{a, b, c}
9	{a, d, e}
10	{b, c}

A. 1/7

B. 1/5

C. 2/5

D. <u>3/5</u>

9. In the following data table, if the support threshold is (greater than or equal to) 0.4 the frequent 3-itemsets are:

Transaction ID	Itemsets
1	{a, b, d, e}
2	{b, c, d}
3	{a, b, d, e}
4	{a, c, d, e}
5	{b, c, d, e}
6	{b, d, c}
7	{c, d}
8	{a, b, c}
9	{a, d, e}
10	{b, c}

A. {a, b, c}

B. {a, b, e}

C. {a, c, e}

D. <u>{a, d, e}</u>

10. Which of the following is an efficient algorithm for finding association rules:

A. Apriori

B. Exhaustive search

C. Bubble sort

D. Insertion sort

Data Mining: Assignment Week 3: Decision Trees

1. The output of a classification algorithm is:

A. Integer of Categorical valued

B. Real or floating point valued

C. Positive valued

D. Negative valued

2. A decision tree consists of:

A. Only leaves

B. Nodes and branches

C. Only a root

D. Nodes connected by cycles

3. Each branch of an internal node in a decision tree corresponds to:

A. A logical condition on an attribute

B. A noisy attribute value

C. A single data instance

D. A null value of the attribute

4. Which of the following criteria is used to decide which attribute to split next in a decision tree:

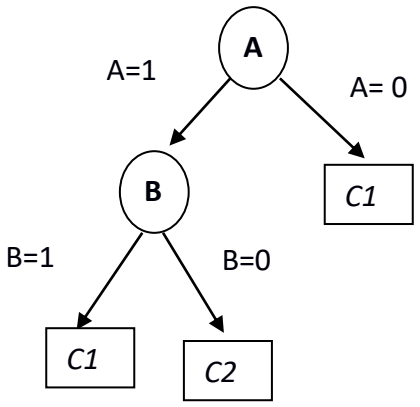
A. Silhouette coefficient

B. Scatter value

C. Information gain

D. Davies-Bouldin index

5. What is the logical rule equivalent to the following decision tree with decision nodes A and B and classes C1 and C2?:



A. (A=0 AND B=1) OR A=0 → *C1*; (A=1 AND B=0) → *C2*

B. (A=1 AND B=0) OR A=0 → *C1*; (A=1 AND B=0) → *C2*

C. (A=1 AND B=1) OR A=0 → *C1*; (A=1 AND B=0) → *C2*

D. (A=1 AND B=1) OR A=0 → *C1*; (A=0 AND B=0) → *C2*

6. If in a data set, the probability of an instance belonging to Class 1 is p_1 and the probability of belonging to Class 2 is p_2. Then the class entropy of the data set is:

A. $-p_1 \log p_1 + p_2 \log p_2$

B. $-p_1 \log p_1 - p_2 \log p_2$

C. $-p_1 \log p_1$

D. $p_2 \log p_2$

8. Given the following training set for classification problem into two classes "email" and "spam". There are two attributes X1 and X2 taking values T or F. Splitting on which attribute in the root of a decision tree will lead to highest information gain?

X1	X2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email

F	F	email
T	T	email
T	F	email

A. X1

B. X2

C. There will be a tie among the attributes

D. Not enough information to decide

8. Given the following training set for classification problem into two classes “email” and “spam”. There are two attributes X1 and X2 taking values T or F. Entropy of the data set before any split is?

X1	X2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email
F	F	email
T	T	email
T	F	email

A. 0

B. **–(4/10)xlog(4/10)-(6/10)xlog(6/10)**

C. –log(4/10)-log(6/10)

D. 1

9. Given the following training set for classification problem into two classes “email” and “spam”. There are two attributes X1 and X2 taking values T or F. Splitting on attribute X1 in the root leads to an entropy reduction of?

X1	X2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email

F	F	email
T	T	email
T	F	email

A. –log(4/10) - log(6/10)

B. **–(4/10)xlog(4/10)-(6/10)xlog(6/10) + (4/7)xlog(4/7) + (3/7)xlog(3/7)**

C. –(4/10)xlog(4/10)-(6/10)xlog(6/10) + (4/7)xlog(4/7) + (3/7)xlog(3/7) + 1

D. 1

10. Given the following training set for classification problem into two classes "email" and "spam". There are two attributes X1 and X2 taking values T or F. Splitting on attribute X2 in the root leads to an entropy reduction of?

X1	X2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email
F	F	email
T	T	email
T	F	email

A. –log(4/10) - log(6/10)

B. –(4/10)log(4/10)-(6/10)log(6/10) + (1/4)log(1/4) + (3/7)log(3/7)

C. <u>–(4/10)log(4/10)-(6/10)log(6/10) + (3/4)log(3/4) + (1/4)log(1/4) + (1/6)log(1/6) + (5/6)log(5/6)</u>

D. 1

11. Generalization performance of a decision tree can be improved using:

A. Increasing height

B. Increasing width

<u>C. Pruning</u>

D. Construction

Data Mining : Assignment Week 4: Bayes Classifier, K-Nearest Neighbor

1. Bayes classifier is based on:

A. Integer values

B. Probability values

C. Floating point values

D. Deterministic values

2. Which of the following classifiers use posterior probability values during classification?

A. Neural network

B. Decision tree

C. Bayes classifier

D. Support vector machine

3. The MAP classifier stands for:

A. Maximum apriori classifier

B. Maximum aposteriori classifier

C. Minimum apriori classifier

D. Minimum aposteriori classifier

4. For events A and C, the Bayes rule states that:

A. $P(A|C)P(C) = P(C|A)P(A)$

B. $P(A|C)P(A) = P(C|A)P(C)$

C. $P(A|C) = P(C|A)P(C)$

D. $P(A|C) = P(C|A)P(A)$

5. A mail system classifies the mails into two classes “email” and “spam” based on their content. We know that the probability of a mail being being spam is *P(spam) = 0.60*, the probability of a mail containing the word “lottery” is *P(lottery) = 0.30*, and the probability that a spam mail contains the word lottery is *P(lottery|spam) = 0.90*. What is the probability of a mail which contains the word lottery being a spam?

A. 0.81

B. 0.50

C. **<u>0.18</u>**

D. 0.10

6. We want to classify a data instance X into either of the two classes C1 or C2. The following probability values are given: P(C1|X) = 0.7, and P(C2|X) = 0.3. What is the class of X as inferred by the Bayes classifier?

<u>A. C1</u>

B C2

C. Both classes equally likely

D. Not enough information to decide

7. Which of the following is true about probabilities of two *independent* attributes A and B?

A. P(A) = P(B)

B. P(A) + P(B) = 1

C. P(A)P(B) = 1

D. <u>P(AB) = P(A)P(B)</u>

8. Consider a binary classification problem with two classes “email” and “spam”. We want to classify a mail *m* into either of the two classes. Class labels of ten other training set instances sorted in increasing order of their distance to the instance *m* is as follows: {email, spam, email, email, email, spam, email, spam, spam, spam}. How will a K=7 nearest neighbor classifier classify *m*?

A. There will be a tie

B. spam

C. email

D. Not enough information to classify

9. Given the following training set for classification problem into two classes “spam” and “email”. There are two attributes X1 and X2 taking values T or F. What is the estimated apriori probability *P(spam)* of the class spam?

A1	A2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email
F	F	email
T	T	email
T	F	email

A. 0.60

B. <u>0.40</u>

C. 0.20

D. 0.10

10. Given the following training set for classification problem into two classes “spam” and “email”. There are two attributes X1 and X2 taking values T or F. What is the estimated class conditional probability *P(X1=T, X2=T|spam)*?

A1	A2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email
F	F	email
T	T	email
T	F	email

A. 0.25

B. 0.35

C. <u>0.75</u>

D. 0.95

11. Given the following training set for classification problem into two classes “spam” and “email”. There are two attributes X1 and X2 taking values T or F. The Bayes classifier classifies the instance (X1=1, X2=1) into class?

A1	A2	Class
T	F	spam
T	T	spam
T	T	spam
T	F	email
T	T	spam
F	F	email
F	F	email
F	F	email
T	T	email
T	F	email

E. <u>spam</u>

F. email

G. there will be a tie

H. not enough information to classify

12. Bayes classifier can be used –

A. Only for two class classification problems

<u>B. Multi-class classification problems</u>

C. Clustering problems

D. Regression problems

Data Mining: Assignment Week 5: Support Vector Machine

1. In classification, the acronym SVM stands for –

A. Super Vector Machine

B. Support Vector Machine

C. Support variable Machine

D. Super Variable Machine

2. SVM is also known as:

A. Maximum Margin Classifier

B. Minimum Margin Classifier

C. Maximum mean Classifier

D. minimum Mean Classifier

3. Margin of a hyperplane is defined as:

A. The angle it makes with the axes

B. The intercept it makes on the axes

C. Perpendicular distance from its closest point

D. Perpendicular distance from origin

4. The equation $W^TX+b=0$, where W is a weight vector and X an input vector, represents a:

A. Hyperplane

B. Sphere

C. Cuboid

D. Ellipsoid

5. Consider a class boundary curve having the equation $W^TX+b=0$. It separates all the training points (X_i, y_i), where class has a value $y_i=\{+1, -1\}$, then:

A. $||W|| = 0$

B. $b = 0$

C. $W^T X_i + b \geq 0$ for all i

D. $y_i(W^T X_i + b) \geq 0$ for all i

6. Consider the following optimization problem in the context of SVMs. The symbols represent their usual values.

Minimize ½ $W^T W$, such that $y_i(W^T X_i + b) \geq 1$ for all i

The solution for W and b, helps us to obtain the:

A. Optimal separating hyperplane

B. Minimum margin classifier

C. Optimal Bayes classifier

D. Zero margin classifier

7. The solutions for the primal and dual optimization problems in the context of SVM are:

A. Always different

B. Always the same

C. Sometimes different

D. Sometimes the same

8. In the solution of the dual optimization problem in the context of SVM, the Lagrange multipliers corresponding to the support vectors have a value:

A. infinity

B. less than zero

C. greater than zero

D. equal to zero

9. The dual optimization problem in SVM design is solved using:

A. Linear programming

B. Quadratic programming

C. Dynamic programming

D. Integer programming

10. The generalization performance of a SVM on the training set and on the unknown test set is controlled by:

A. Dimensionality of input data

B. Number of support vectors

C. Slack in the soft margin

D. Generalization constant C

11. Let X_j be a support vector of a hard margin SVM, we have:

A. $|W^T X_j + b| = 0$

B. $|W^T X_j + b| < 1$

C. $|W^T X_j + b| = 1$

D. None of the above

Data Mining: Assignment Week 6: ANN

1. The class boundary in a perceptron classifier is a:

A. Circle

B. Parabola

C. Straight line

D. Ellipse

2. The number of neurons in a single perceptron classifier is?

A. 4

B. 3

C. 2

D. 1

3. The MLP stands for:

A. Multi-Layered Perceptron

B. Mini-Layered Perceptron

C. Multi-Labelled Perceptron

D. Mini-Labelled Perceptron

4. Which of the following logic functions can be implemented by a MLP, but cannot be implemented by a single perceptron?

A. AND

B. OR

C. NOR

D. XOR

5. An MLP with a single hidden layer has 3 nodes in the input layer, 2 nodes in the hidden layer, and 1 node in the output layer. The total number of weights of the fully connected MLP is:

A. 8

B. 6

C. 4

D. 2

6. Which of the following is NOT true about the MLP?

A. It is a feed-forward network

B. It is a fully connected network

C. It is a neural network

D. It is a feed-back network

7. A sigmoid activation function is commonly used in the MLP. The sigmoid activation function $h(z) = \frac{1}{1+e^{-z}}$ is:

A. Discontinuous and not differentiable

B. Discontinuous but differentiable

C. Continuous but not differentiable

D. Continuous and differentiable

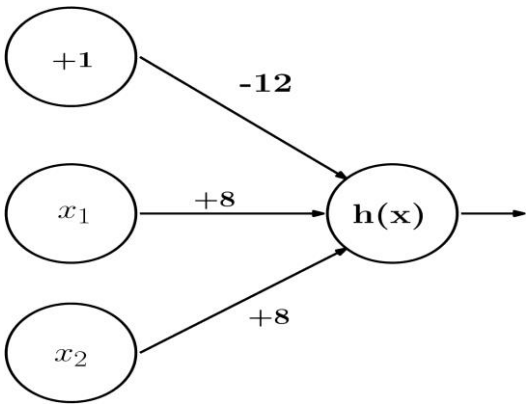
8. Which of the following step is used to update the weights of an MLP during the training phase:

A. Distillation
B. Backpropagation
C. Discretization
D. Elimination

9. Which of the following optimization technique is implemented by backpropagation during neural network training?

A. Quadratic programming
B. Linear programming
C. Gradient descent
D. Evolutionary algorithms

10. Consider the neural network given below. It takes two binary valued inputs $x_1, x_2 \epsilon$ {0,1} and the activation function is the binary threshold function ($h(x) = 1 \; if \; x > 0; 0 \; otherwise$). What is the output of the neural network for the input vector [1, 1]?



A. 0

B. 1

C. 4

D. 16

11. When do you expect overfitting to happen in the following scenarios of neural network training iterations?

A. training set as well as test set error decreases

B**. training set error decreases but test set error increases**

C. training set error as well as test set error increases

D. training set as well as test set error remains constant

Data Mining: Assignment Week 7: Clustering

1. Which of the following is an unsupervised learning algorithm?

A. Two class classification

B. Multiclass classification

C. Clustering

D. Regression

2. A cluster refers to a –

A. group of similar data items

B. group of dissimilar data items

C. outlier data items

D. incomplete data items

3. The value of Euclidean distance between the pair of two-dimensional points [0, 0] and [1, 1] is approximately–

A. 0.000

B. 1.000

C. 1.414

D. 2.404

4. Consider a set of five 2-dimensional points $p_1=(0, 0)$, $p_2=(10, 5)$, $p_3=(10, 7)$, $p_4=(0, 1)$, and $p_5=(0, 2)$. Euclidean distance is the distance function and scatter coefficient is the cluster quality measure. Which among the following groupings denotes the best clustering of the data points?

A. $\{p_1, p_2, p_3\}$ $\{p_4, p_5\}$

B. $\{p_1, p_2, p_5\}$ $\{p_3, p_4\}$

C. $\{p_1, p_2, p_4\}$ $\{p_3, p_5\}$

D. $\{p_1, p_4, p_5\}$ $\{p_2, p_3\}$

5. Which of the following is NOT a clustering algorithm?

A. K-means

B. Single linkage

C. DBSCAN

D. K-nearest neighbour

6. Which of the following is not a hierarchical clustering algorithm?

A. K-Means

B. Single linkage

C. Complete linkage

D. Average linkage

7. Which of the following is NOT true about the K-Means clustering algorithm?

A. It tends to produce spherical clusters

B. The final solution depends on the initial choice of means

C. It is computationally the most expensive clustering algorithm

D. It is an iterative clustering algorithm

8. The tree used to obtain clusters in a hierarchical clustering algorithm is known as:

A. Histogram

B. Dendrogram

C. Parallelogram

D. Unigram

9. Consider a set of five 2-dimensional points $p_1=(0, 0)$, $p_2=(10, 5)$, $p_3=(10, 7)$, $p_4=(0, 1)$, and $p_5=(0, 2)$. Euclidean distance is the distance function. Single linkage clustering is used to cluster the points into two clusters. The clusters are:

A. $\{p_1, p_2, p_3\}$ $\{p_4, p_5\}$

B. $\{p_1, p_4, p_5\}$ $\{p_2, p_3\}$

C. $\{p_1, p_2, p_5\}$ $\{p_3, p_4\}$

D. $\{p_1, p_2, p_4\}$ $\{p_3, p_5\}$

10. Consider a set of five 2-dimensional points $p_1=(0, 0)$, $p_2=(10, 5)$, $p_3=(10, 7)$, $p_4=(0, 1)$, and $p_5=(0, 2)$. Euclidean distance is the distance function. The k-means algorithm is used to cluster the points into two clusters. The initial cluster centers are p_1 and p_3. The clusters after one iteration of k-means are:

A. $\{p_1, p_4, p_5\}$ $\{p_2, p_3\}$

B. $\{p_1, p_2, p_3\}$ $\{p_4, p_5\}$

C. $\{p_3, p_4, p_5\}$ $\{p_1, p_2\}$

D. $\{p_1, p_2, p_4\}$ $\{p_3, p_5\}$

11. Which of the following is an example of density-based clustering algorithm?

A. DBSCAN

B. K-means

C. Single linkage

D. Complete linkage

Data Mining: Assignment Week 8: Regression

1. Regression is an example of –

A. Supervised learning

B. Unsupervised learning

C. Clustering

D. Classification

2. The output of a regression function is usually –

A. Categorical valued

B. Real valued

C. Negative valued

D. Positive valued

3. If x_1, x_2 are independent variables and y the dependent variable, which of the following represents a linear regression model?

A. $y = a_0 + a_1/x_1 + a_2/x_2$

B. $y = a_0 + a_1x_1 + a_2x_2$

C. $y = a_0 + a_1x_1 + a_2x_2^2$

D. $y = a_0 + a_1x_1^2 + a_2x_2$

4. The output of a regression model is the –

A. Independent variable

B. Dependent variable

C. Integer variable

D. Null variable

5. Which of the following is most commonly used as an error measure in linear regression?

A. Mean squared error

B. Classification error

C. Scatter coefficient

D. Silhouette coefficient

6. The number of input variables in linear regression can be -

A. One

B. Two

C. Three

D. Many

7. The linear regression model $y = a_0 + a_1x$ is applied to the data in the table shown below. What is the value of the mean squared error, when $a_0 = 0$, $a_1 = 1$?

x	y
0	1
1	2
2	3
2.5	3.5

A. 0.00

B. 0.25

C. 0.50

D. 1.00

8. The linear regression model $y = a_0 + a_1x$ is to be fitted to the data in the table shown below. What is the optimal regression model obtained by minimizing sum squared error?

x	y
0	1
1	2
2	3
2.5	3.5

A. $y = 1.01 – 2.10x$

B. $y = 1.00 + 2.10x$

C. $y = 1.01 – 1.00x$

D. $y = 1.00 + 1.00x$

9. The linear regression model $y = a_0 + a_1x_1 + a_2x_2$ is applied to the data in the table shown below. What is the value of the mean squared error, when $a_0 = 0$, $a_1 = 1$, $a_2 = 1$?

x_1	x_2	y
0	0	1
0	1	2
1	0	2
1	1	3

A. 0.00

B. 0.25

C. 0.50

D. 1.00

10. We consider Regression Model A to be a linear regression line, and Regression Model B to be a cubic regression curve. Which of the following statement is true when these models are trained on a small data set?

A. Model A might underfit, and Model B might overfit

B. Model A might overfit, and Model B might underfit

C. Model A is more complex than Model B

D. Model A will have higher training set error than Model B

11. Autoregressive models for a time series predict –

A. Values of future time steps from past time step values
B. Values of past time steps from future time step value
C. Values of present time step present time step value
D. Values of present time step from a different regression model